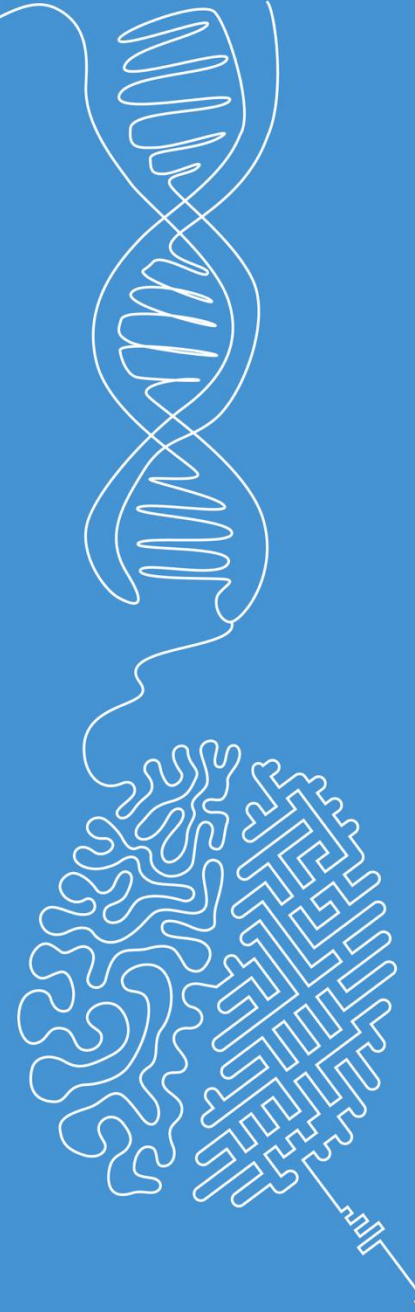


Filtering II

Image and Signal Processing

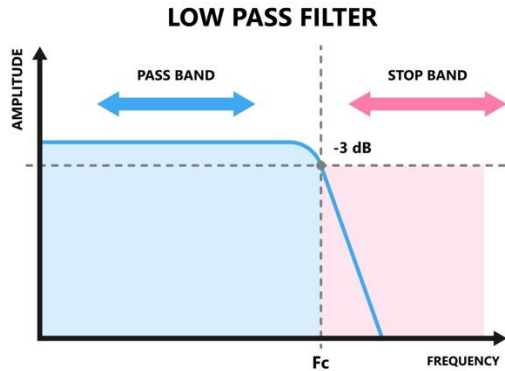
Norman Juchler



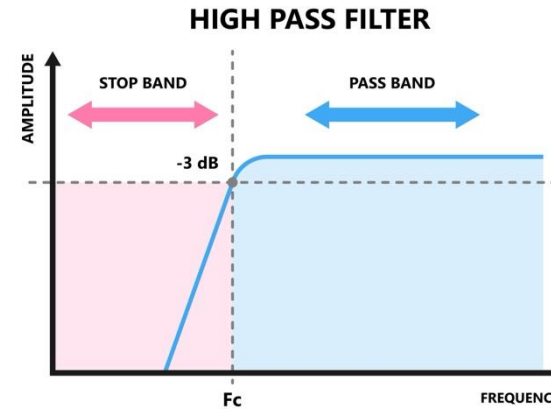
Common filter types

...and their characteristics

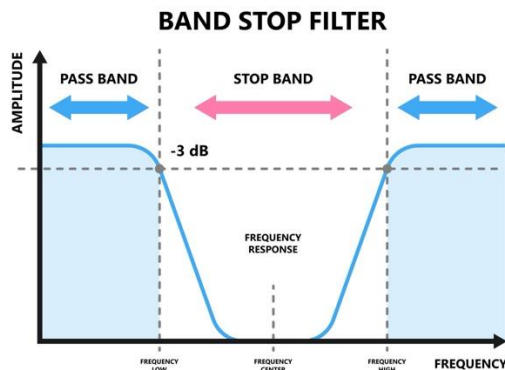
Overview: Types of filters



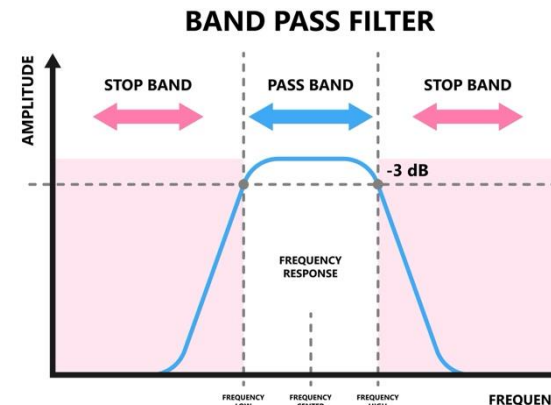
- Allows frequencies below a cutoff frequency to pass through while attenuating higher frequencies.
- Used to remove high-frequency noise, smooth signals, or extract low-frequency components.



- Allows frequencies above a cutoff frequency to pass through while attenuating lower frequencies.
- Used to remove low-frequency noise, extract high-frequency components, or emphasize high-frequency features.



- Attenuates frequencies within a specific range, known as the stopband, while allowing frequencies outside this range to pass through.
- Used to remove unwanted interference or noise within a certain frequency range while preserving the rest of the signal.

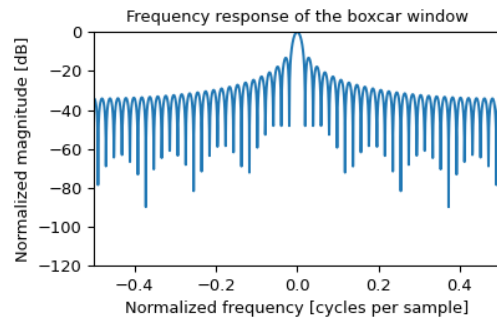
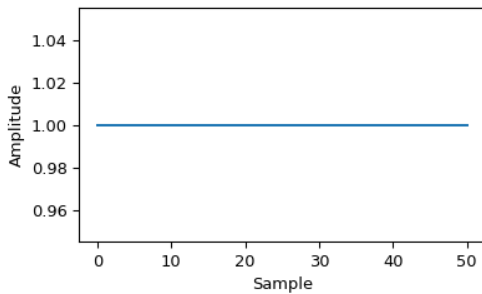


- Allows a specific range of frequencies, known as the passband, to pass through while attenuating frequencies outside this range.
- Used to isolate or extract signals within a certain frequency band while rejecting others.

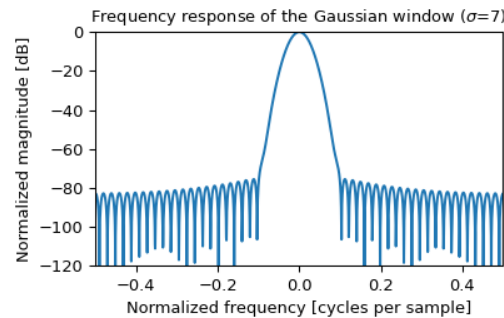
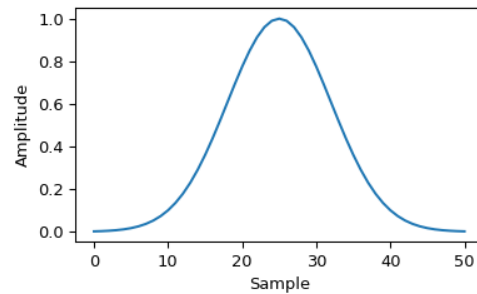
Filters have properties in the time- and frequency domain

Comparison of different window filters. Different window functions lead to different time–frequency trade-offs. Narrow windows preserve detail, while smooth windows reduce spectral artifacts.

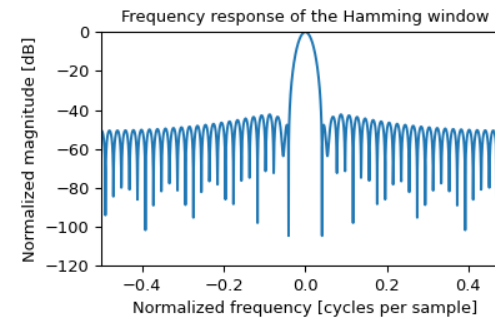
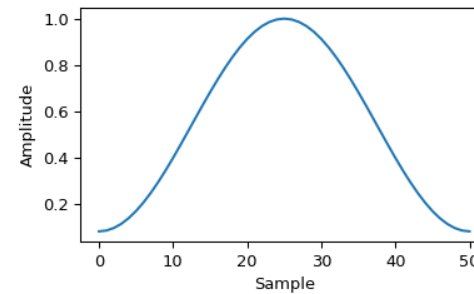
Boxcar window



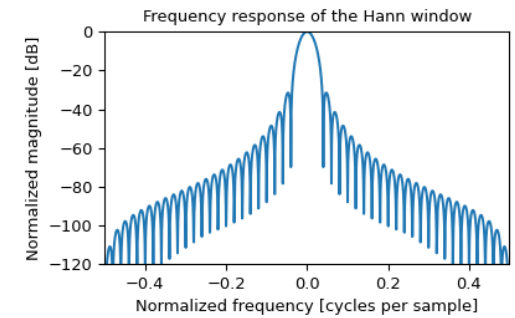
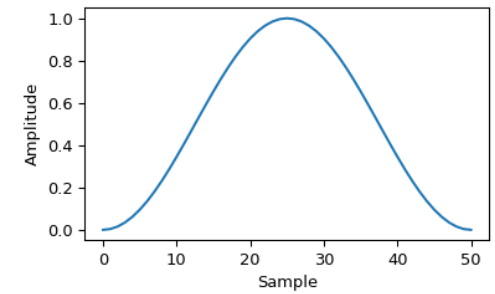
Gaussian window ($\sigma=7$)



Hamming window



Hann window



$$h_{box}[n] = 1$$

$$h_{gauss}[n] = e^{\frac{1}{2}\left(\frac{n}{\sigma}\right)^2}$$

$$h_{lanc}[n] = \text{sinc}_{\pi}\left(\frac{2n}{M-1} - 1\right) \quad h_{hann}[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right)$$

$$w_{box}[n] = \begin{cases} 1, & 0 \leq n < N \\ 0, & \text{otherwise} \end{cases}$$

Common Filtering Methods: FIR vs. IIR

FIR and IIR filters are implemented in the time domain, but their effects are best understood in the frequency domain.

■ FIR Filtering (Finite Impulse Response)

- Impulse response has finite duration (i.e., decays to zero in finite time).
- Output depends only on current and past inputs (no feedback).
- Always stable: a bounded input always produces a bounded output.
- Can achieve linear phase, preserving the shape of the signal.
- Typically requires more coefficients for sharp filters
→ higher computational cost.

A FIR filter has the following form, where $x[n]$ is the input signal and b_k are the filter coefficients

$$y[n] = \sum_{k=0}^M b_k \cdot x[n - k]$$

■ IIR Filtering (Infinite Impulse Response)

- Impulse response is infinite due to feedback.
- Output depends on past inputs and past outputs.
- May become unstable if not carefully designed.
- Can achieve sharp filtering with fewer coefficients
→ more efficient.

A IIR filter has the following form, where $x[n]$ is the input signal, b_k are the feedforward coefficients, and a_k are the feedback coefficients

$$y[n] = \sum_{k=0}^M b_k \cdot x[n - k] - \sum_{k=1}^N a_k \cdot y[n - k]$$

Common filter types

We touch on this topic only briefly. As future engineers, you should have at least heard of these filter types. Filter design – an engineering discipline – involves selecting parameters to meet specific output requirements.

■ Butterworth filter:

- Maximally flat in the passband, providing a smooth response without ripples, but with a relatively gradual cutoff.

■ Chebyshev filters:

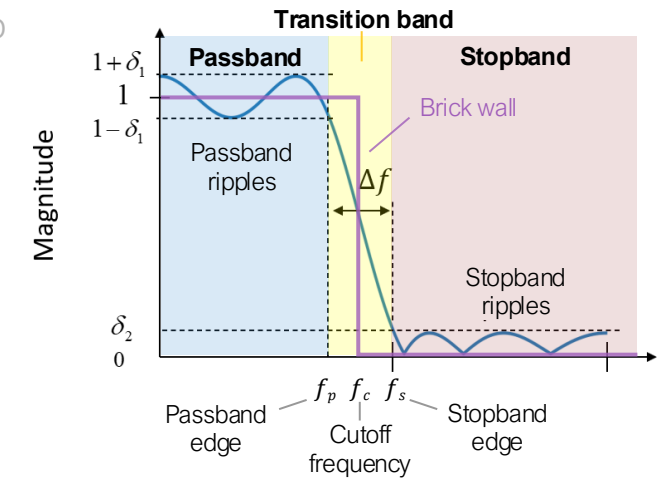
- Offer a sharper transition between passband and stopband...
- ...at the cost of ripples (Type I: passband ripple, Type II: stopband ripple).

■ Windowed FIR filters:

- Designed by applying a window function (e.g., Hamming, Hann, Blackman) to an ideal filter, controlling the trade-off between main-lobe width and side-lobe suppression.

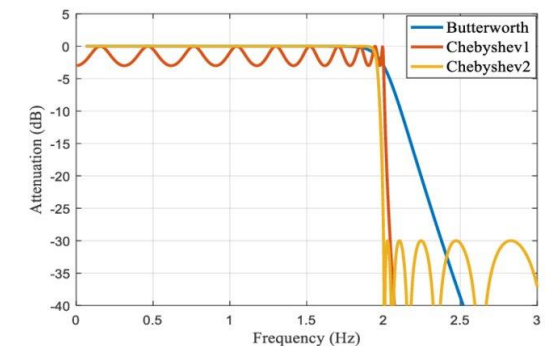
■ Adaptive filters:

- Continuously adjust their coefficients based on the input signal (e.g., LMS algorithm); commonly used in applications such as noise cancellation and echo suppression.



Filter specification: Graphical illustration of the amplitude response of a low-pass filter, highlighting the key design parameters.

Adapted from [Dan Boschen / DSP Stack](#)



Amplitude responses of a Butterworth and two variants of the Chebyshev filter. [Source](#)

Example: Butterworth filter

- A common type of filter used in signal processing
- Can be implemented as an analog or digital filter
- **Design goal:** Achieve a frequency response that is as flat as possible in the passband
- Also known as a “maximally flat magnitude filter”
- Typically implemented as an **Infinite Impulse Response (IIR)** filter
- Can be modeled as a **Linear Time-Invariant (LTI) system**: Example:

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1}{1 + 2s + 2s^2 + s^3}$$

